Agentic Economy on Arc Hackathon

The Agent Economy Built for Sub-Cent Transactions

Usage-Based Compute Billing

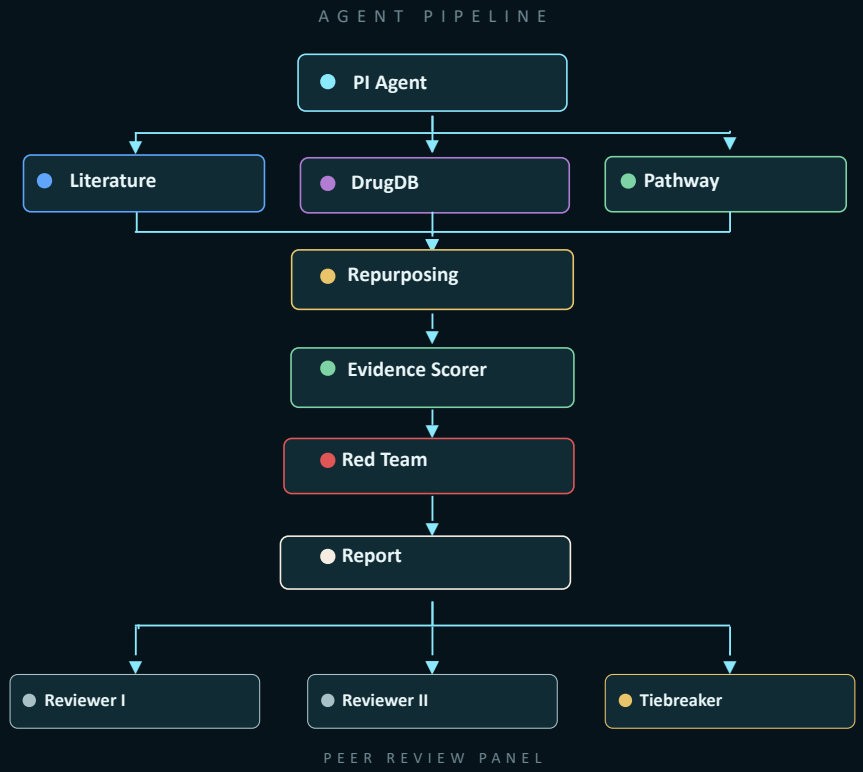
Real-Time Micro-Commerce Flow

BioMed Research Engine

Veliora

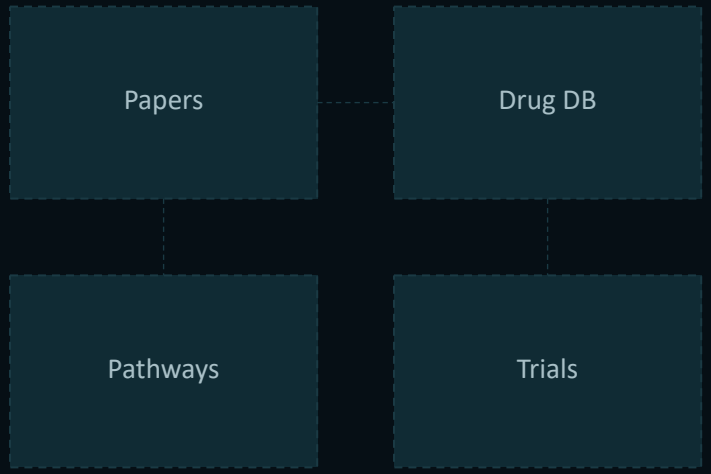
Agentic. Structured. Usage-based.

- Disease-focused discovery workflows
- Specialist agents, structured outputs
- Usage-based execution on Arc + Circle



Biomedical research is too fragmented and too information-dense

- Evidence is spread across papers, drug databases, pathway resources, and trial records
- Existing tools improve parts of the process, but usually not the full workflow
- Small teams still need to connect these layers into one structured and traceable research output



33M+

biomedical records
in PubMed

1 in 3

PubMed searches
return 100+ results

Nature '25

fragmented tools remain
major workflow bottlenecks

Sources: PubMed / Oxford Academic / Nature Scientific Reports

Veliora turns scattered research steps into one structured workflow

From disease question to validated research brief

RESEARCH GATHERING

PI Agent → Literature → Drug DB

▼

SYNTHESIS & EVALUATION

Pathway → Repurposing → Evidence Scorer

▼

FINAL OUTPUT

Red Team → Report → Peer Review

Who this is for

PRIMARY

- Early-stage biotech teams
- Translational research labs

SECONDARY

- Rare disease foundations
- Independent scientific consultants

Rare disease is a strong wedge

300M

people affected globally

~95%

lack approved treatments

Sparse evidence, limited expert capacity

Structured synthesis is most valuable when evidence is thin and capacity is scarce

Sources: The Lancet Global Health / NIH NCATS / NCBI PMC

Competitive Landscape

Product	Category	Pathway	Drug/ Target	Multi-step Agents	Pricing
Elicit	Lit. search	✗	✗	✗	$0–49/mo
Consensus	Lit. synthesis	✗	✗	✗	$0–15/mo
Perplexity	General search	✗	✗	✗	$0–20/mo
PandaOmics	Target ID, pathways, agents	✓	✓	✓ PandaClaw (Mar 2026)	No public pricing
Recursion OS	Full drug discovery	✓	✓	✓	No public pricing
BenevolentAI	Drug discovery (pharma only)	✓	✓	✗ SaaS discontinued	No public pricing
BioMed Research	Disease-focused pipeline	✓	✓	✓	Usage-based / per-task

U S P

- Specialist agents
- Traceable outputs
- Evidence scoring
- Built-in critique
- Per-task execution

Sources: Official pricing pages / official product pages

Arc + Circle make the model viable

- Paid service stages run through x402
- 50 paid actions at $0.002 each
- Traditional onchain settlement is not viable at that price point
- Circle batches settlement
- Arc provides stable, USDC-native settlement

ECONOMICS

$0.002

per agent action

$0.01+

typical onchain tx cost

5x below viable threshold for traditional settlement

Real workflow. Real payments. Real completion.

10	50	$0.002	E2E
Completed Runs	Paid Specialist Actions	Avg Price Per Action	Lifecycle Tx Tracked

Transactions verified via Arc Explorer | Live proof available

Sources: Live proof / transaction summary / Arc explorer

How the workflow runs

1. Create a research request
2. Fund the task onchain
3. PI agent assigns work and triggers paid x402 calls
4. Agents produce the final report
5. Peer review finalizes the task

What the final report includes

- Summary & Methodology
- Validation Checks & Trials
- Genetic Evidence & Novelty Search
- Failure Patterns & Hypotheses
- Discovery Candidates & Evidence

Veliora makes biomedical research

more structured

more traceable

more accessible

Agentic biomedical research with per-task execution

Tech Stack

Layer	Technology
Frontend	Next.js, React, TypeScript
Backend	Node.js, Express, TypeScript
State	SQLite, workflow event logging
Payments	Arc, Circle, x402, ERC-8183
Models	Gemini, GPT-4o, Qwen

Multi-model reasoning with programmable payment and settlement infrastructure